Electric bench scissor lift table, 150 kg, 250-900 mm

ScissorLift - calculation report. Designed and calculated only: nothing has been built or tested. CAD = measured in Inventor, CALC = calculated, CAT = catalogue / vendor listing.

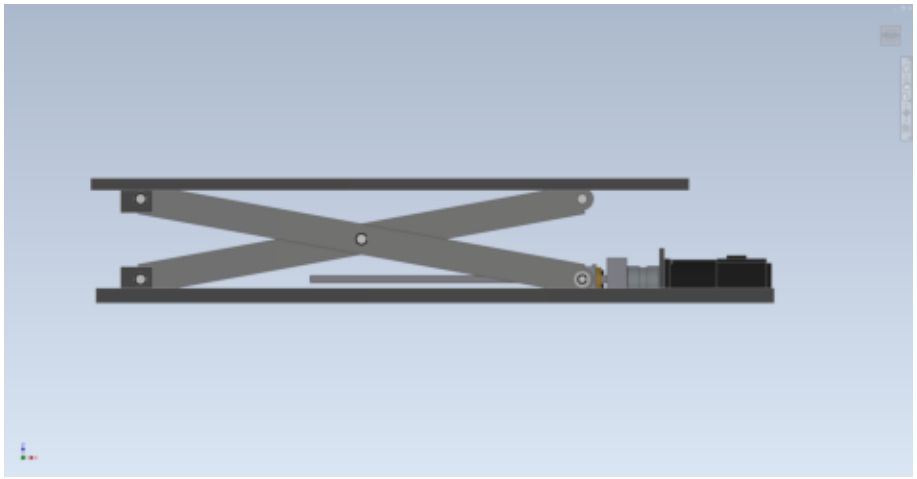


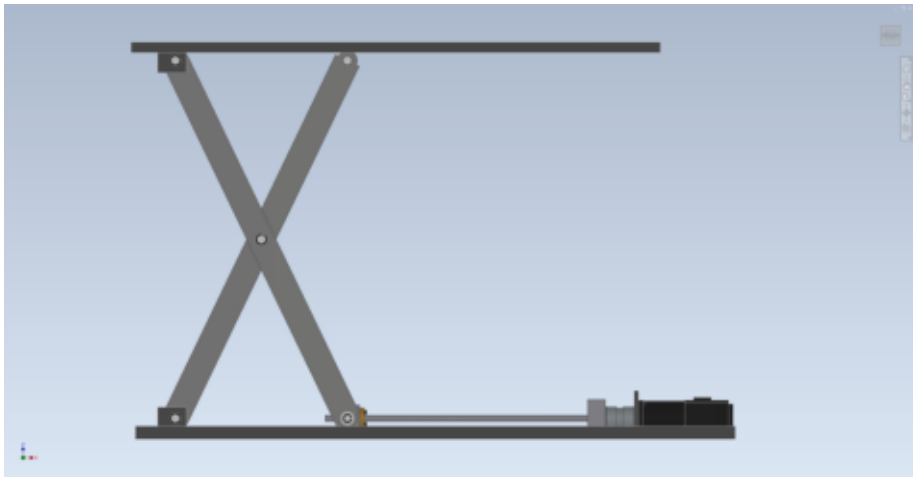
Item	Value	Source
Rated load	150 kg on a 1200 x 560 deck	spec
Height	250 collapsed to 900 mm; arm angle 10.3 to 64.3 deg	CAD
Mechanism	single scissor, 900 pin centres, RHS 60x30x3 S355 arms, 20 mm pins in composite bushings	CAD
Drive	TR16x4 screw pushing a nut carriage on the roller axis; StepperOnline 24 V 200 W BLDC + 71.5:1 (33 Nm, 42 rpm)	CAT
Screw force	10.72 kN collapsed (incl. 2.6 % friction), 0.91 kN raised	CALC
Motor margin	SF 2.20 (mu 0.10) / 1.73 (mu 0.15) on rated torque	CALC
Lift time	177 s for 495 mm of carriage travel (124 screw turns)	CALC
Weakest part	carriage roller pin SF 3.15; arms SF 5.0; thrust bearing s0 1.73	CALC
Mass	80.8 kg (platform 32.2 kg)	CAD
Assembly	76 occurrences, 225 constraints, 0 DOF; 0 clashes at 7 heights	CAD
Cost	about $1373 incl. controls (estimates, see BOM)	est.

1. Geometry and kinematics

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Item	Value	Source
Deck height	H = z_b + L sin(theta) + top = 47.5 + 900 sin(theta) + 41.5 = 89 + 900 sin(theta)	CALC
Stack heights	z_b = base rail 30 + roller radius 17.5; top = roller 17.5 + rail 20 + deck 4 (sets the 250 mm collapsed height)	CAD
Arm angle	theta = asin((H - 89)/900): 10.31 deg at 250, 64.30 deg at 900	CALC
Carriage position	x = 900 cos(theta): 885.5 mm to 390.2 mm, travel 495.3 mm	CALC
Why the screw drives the rollers	the screw sits on the carriage pivot axis (z 47.5), so its force puts no moment on the nut; the carriage is pushed toward the fixed pivots, the screw is in compression and short when the force is high	design
Inventor driving parameter	Lift (mm); theta = asin((Lift - 89 mm)/900 mm); platform flush = Lift - 41.5 mm; carriage flush = 900 mm cos(theta); arm angle constraints = theta / -theta	CAD
Kinematics in the assembly	deck height, arm angle and carriage X measured at 7 heights: max error 0.0000 mm / 0.00000 deg / 0.0000 mm	CAD
Envelope	1370 x 560 mm footprint (gearmotor at the far end)	CAD

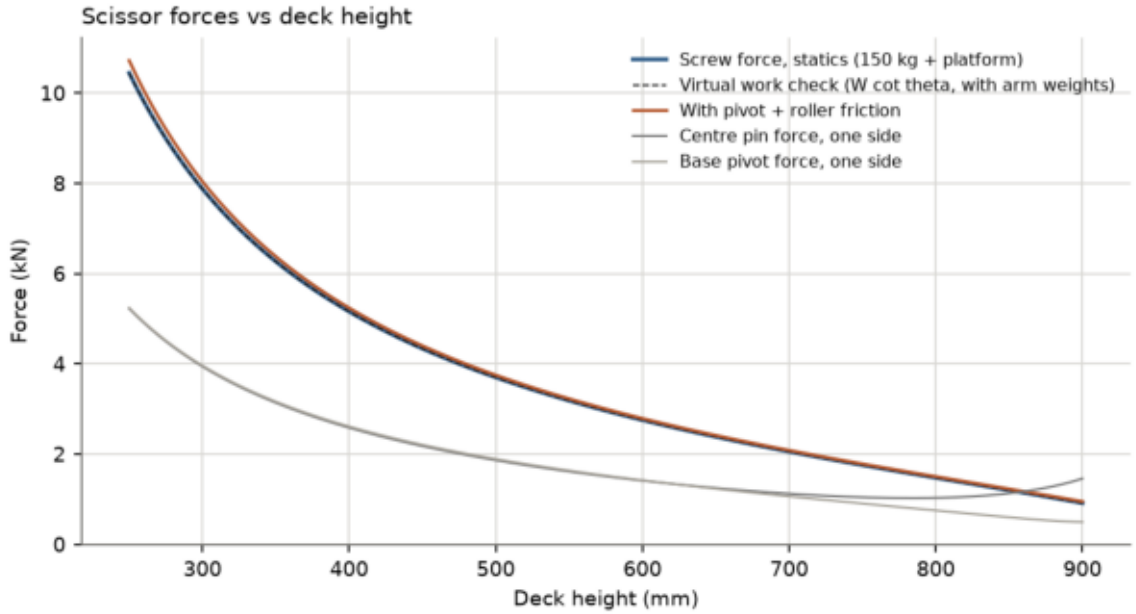




2. Scissor statics and screw force

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Each arm side is solved as four rigid bodies (platform, inner arm, outer arm, nut carriage) with 11 unknowns: pin forces at the base pivot B, platform pivot A, centre pin E and carriage pin D, the vertical roller reaction C under the platform rail, the screw share S and the carriage roller reaction N. Masses and CGs are Inventor's. Load cases (EN 1570-1 style): rated load over the full deck, over the far half, over the pivot half, and over one lateral half (the near arm pair then takes 87.6 %). The screw force from statics matches virtual work, S = (W + platform) cot(theta) + arm weights, to 3.1e-15. Pivot friction (mu 0.10 in the bushings, centre pin turning at twice the arm rate) and roller drag (0.5 %) are added by virtual work.

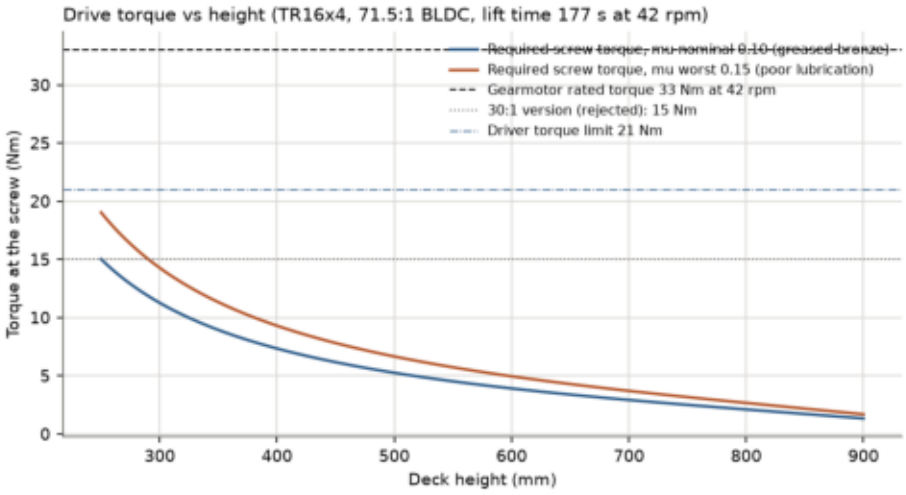


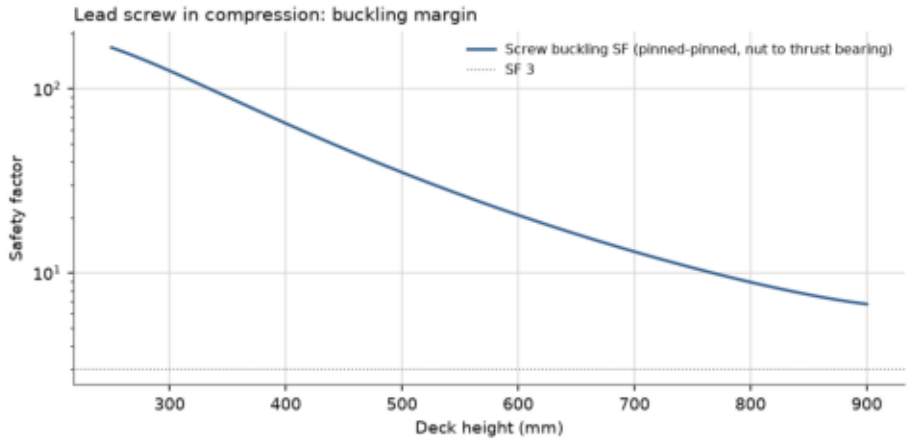
Joint	Peak force	Case	H mm
Base pivot (inner arm)	8.95 kN	Half deck, one side	250
Platform pivot (outer arm)	0.81 kN	Half deck, far end	900
Centre pin	8.92 kN	Half deck, one side	250
Carriage roller pin (outer arm)	8.97 kN	Half deck, one side	250
Screw (total)	10.72 kN	any (independent of load position)	250
Track roller (one)	2.06 kN	half deck, far end	900

3. Drive: lead screw, nut and gearmotor

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Item	Result	Source
Screw	TR16x4 single start, d2 14, lead angle 5.20 deg; efficiency 0.463 (mu 0.10) / 0.364 (mu 0.15)	CALC
Self-locking	locks while mu > tan(lambda) cos(15) = 0.088. Greased bronze static mu is 0.10-0.15, so it should hold at rest, but the margin is small: vibration can make it creep. Spec a brake or the maintenance prop (open item).	CALC
Torque at the screw	peak 15.02 Nm (mu 0.10) / 19.04 Nm (mu 0.15) at 250 mm; 7.3 Nm at 400, 1.3 Nm at 900; RMS over a lift 4.3 / 5.5 Nm	CALC
Gearmotor	M82B200-24P-30S/G82-71.5S1: 33 Nm rated at 42 rpm -> SF 2.20 / 1.73	CAT/CALC
Rejected: 30:1 version	15 Nm rated: SF 1.00 / 0.79, so it stalls at the bottom with poor lubrication; it would lift in 74 s	CALC
Lift time	124 turns at 42 rpm = 177 s; peak shaft power 66 W of 200 W	CALC
Lowering	self-locking screw: the motor drives down, 0.91 Nm at 250 mm	CALC
Driver torque limit	set the current limit to 21 Nm at the screw (above the 19.0 Nm worst need) so a jam cannot overload the 10 mm journal or the coupling	design
Nut	bronze flange nut, 40 long = 10 threads: p = 12.2 MPa (allow ~15 MPa, slow, intermittent); sliding 31 mm/s, pv 0.38 MPa m/s	CALC
Screw stress	compression 103 MPa + torsion 64 MPa at the root -> von Mises 151 MPa, SF 2.25 (C45, Re 340)	CALC
Journal 10 mm	107 MPa at the 21 Nm limit (clamp hub, no keyway): SF 1.83 on shear yield	CALC
Coupling	Lovejoy L-095 NBR 21.9 Nm nominal: SF 1.15 on the worst running torque (jaw couplings take ~2x nominal peaks)	CAT/CALC
Screw buckling	pinned-pinned, nut to thrust bearing: minimum SF 6.8 (at 900 mm, where the force is lowest)	CALC
Thrust bearing 51200	C0 18.6 kN: s0 1.73; cubic-mean load 3672 N, L10 33 years at 20 lifts/day	CAT/CALC





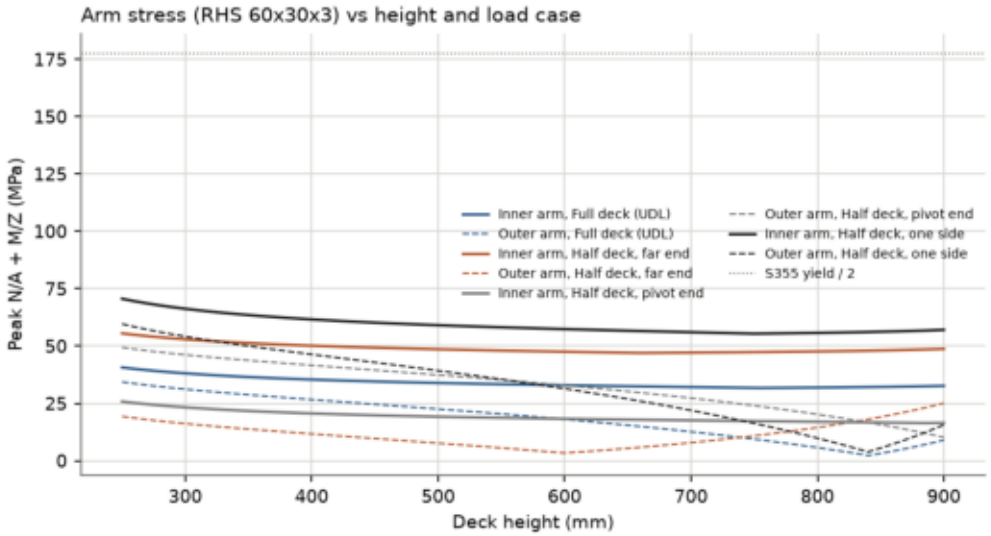
4. Pins, bushings and arms

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Pin (20 mm 42CrMo4, Re 650)	F N	M Nm	vM MPa	SF	Bushing p MPa	Plate / block bearing MPa
Base pivot (inner arm)	8953	60.4	81	8.04	14.9	28
Platform pivot (outer arm)	810	5.5	7	88.97	1.4	2
Centre pin	8924	71.4	103	6.29	14.9	-
Carriage roller pin (outer arm)	8971	157.2	206	3.15	14.9	18

Clevis pins: simply supported between the plate centres with the eye load spread over 30 mm, M = F((b + 2g + a)/4 - b/8). Centre pin: two adjacent eyes, M = F(b + g)/4. Carriage roller pin: cantilever from the end block, outer-arm force at the boss centre plus the roller load at the roller centre. SKF PCM bushings allow 250 MPa static / 80 MPa dynamic; the highest pressure is 14.9 MPa and pv at the centre pin is 0.0016 MPa m/s (limit 1.6).

Arms	Result	Source
Section	RHS 60x30x3: A 504 mm2, Z 7502 mm3, net at the centre bore 7300 mm3	CALC
Peak stress	70.6 MPa (Half deck, one side / I, collapsed, at the centre pin): SF 5.03 on S355	CALC
Column buckling	max compression 8.9 kN vs Euler 186 kN (weak axis, 900 pin-pin): SF 21	CALC
Fatigue (welds)	FAT 71 detail, 50000 cycles (20 lifts/day, 10 years), partial factor 1.35: allow 180 MPa range, SF 2.55 taking the full peak as the range	CALC



5. Platform, rollers, stability and the CAD checks

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Item	Result	Source
Deck frame	4 mm plate + 4 RHS 40x20x2: I 152661 mm4, Z 8220 mm3 (bottom fibre)	CALC
Point load 150 kg	mid-span collapsed 326 Nm, 0.66 mm; at the far edge when raised (610 mm past the roller) 897 Nm -> 109 MPa, SF 2.15, tip 5.7 mm	CALC
Track rollers 6202-2RS	peak 2061 N per roller vs C0 3.75 kN: s0 1.82 (INA LR202-2RSR crowned track roller recommended)	CAT/CALC
Tip-over, lateral	rated load on one lateral half at 900 mm plus a 147 N push (10 % of rated) at deck height: SF 3.04	CALC
Tip-over, longitudinal	rated load on the pivot half, same push along X: SF 6.87	CALC
Corner load to tip	base rails reach y = 269 of the 280 deck half-width: a corner load would need 19.4 kN to lift a rail	CALC

Lift mm	Arm deg	Carriage x mm	Clashing pairs	Inner-outer frame mm	Carriage-drive block mr	Inner frame-screw mm
250	10.30	885.5	0	2.0	21.5	33.3
300	13.56	874.9	0	2.0	32.1	52.6
400	20.22	844.6	0	2.0	62.4	92.0
500	27.17	800.7	0	2.0	106.3	132.8
650	38.56	703.8	0	2.0	203.2	198.4
800	52.19	551.8	0	2.0	355.2	274.4
900	64.30	390.2	0	2.0	516.8	312.6

Inventor: 76 occurrences, 23 files, 225 constraints {'Flush': 169, 'Mate': 54, 'Angle': 2}, total DOF 0, unhealthy 0.

6. Assumptions, limits and open items

Assumptions
- Rated load 150 kg as a uniformly distributed load on the four EN 1570-1 style areas; no dynamic factor on the stresses (slow lift, 2-6 mm/s deck speed).
- Friction: bronze nut on a greased steel screw mu 0.10 nominal, 0.15 worst; composite bushings mu 0.10; roller drag 0.5 %; thrust ball bearing mu 0.003.
- Gearmotor data (33 Nm rated, 42 rpm, 200 W) are from the StepperOnline listing as shown in search results; flange, shaft and lengths are an envelope.
- Steel: S355J2H arms (Re 355), S235 plate frames (Re 235), 42CrMo4 +QT pins (Re 650), C45 screw (Re 340, conservative for rolled +N).
- Bearing ratings: 6202 C 8.06 / C0 3.75 kN, 6200 C 5.07 / C0 2.36 kN, 51200 C 12.7 / C0 18.6 kN (distributor listings of SKF data).
- Tip-over push 10 % of rated load is my assumption for a hand push on the deck; the base sits on a level floor and is not anchored.
- Beam theory, no FEA: the highest stresses are 71 MPa in the arms and 206 MPa in a 42CrMo4 pin, all simple shapes.

Open items (to resolve before building)
1. Self-locking margin: TR16x4 holds only while mu > 0.088. Add a spring-applied brake (a braked BLDC or a brake on the screw) or always use the maintenance prop.
2. Confirm the gearmotor drawing (flange, shaft, length) and price; the motor plate holes and the coupling bore follow an assumed 82 mm flange and 16 mm shaft.
3. Nut pressure 12.2 MPa at the start is near the ~15 MPa bronze guideline: a 50 mm long nut would bring it to 9.8 MPa if the vendor offers one.
4. Track rollers: replace the plain 6202-2RS by INA LR202-2RSR crowned track rollers (same envelope) for edge loading on the rail.
5. Prices are estimates: no vendor page returned a readable price on 2026-10-29.
6. Not modelled: driver, power supply, pendant, limit switches, E-stop, cable chain, coupling guard, maintenance prop (in the BOM as estimates).